



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Experiment 2: Finite State Automaton (Strings Ending with 'ab')

Program

```
def ends_with_ab(string):
```

```
    state = 0
```

```
    for ch in string:
```

```
        if state == 0:
```

```
            if ch == 'a':
```

```
                state = 1
```

```
            else:
```

```
                state = 0
```

```
        elif state == 1:
```

```
            if ch == 'b':
```

```
                state = 2
```

```
            elif ch == 'a':
```

```
                state = 1
```

```
            else:
```

```
                state = 0
```

```
        elif state == 2:
```

```
            if ch == 'a':
```

```

        state = 1

    else:

        state = 0

    return state == 2

s = input("Enter a string: ")

if ends_with_ab(s):

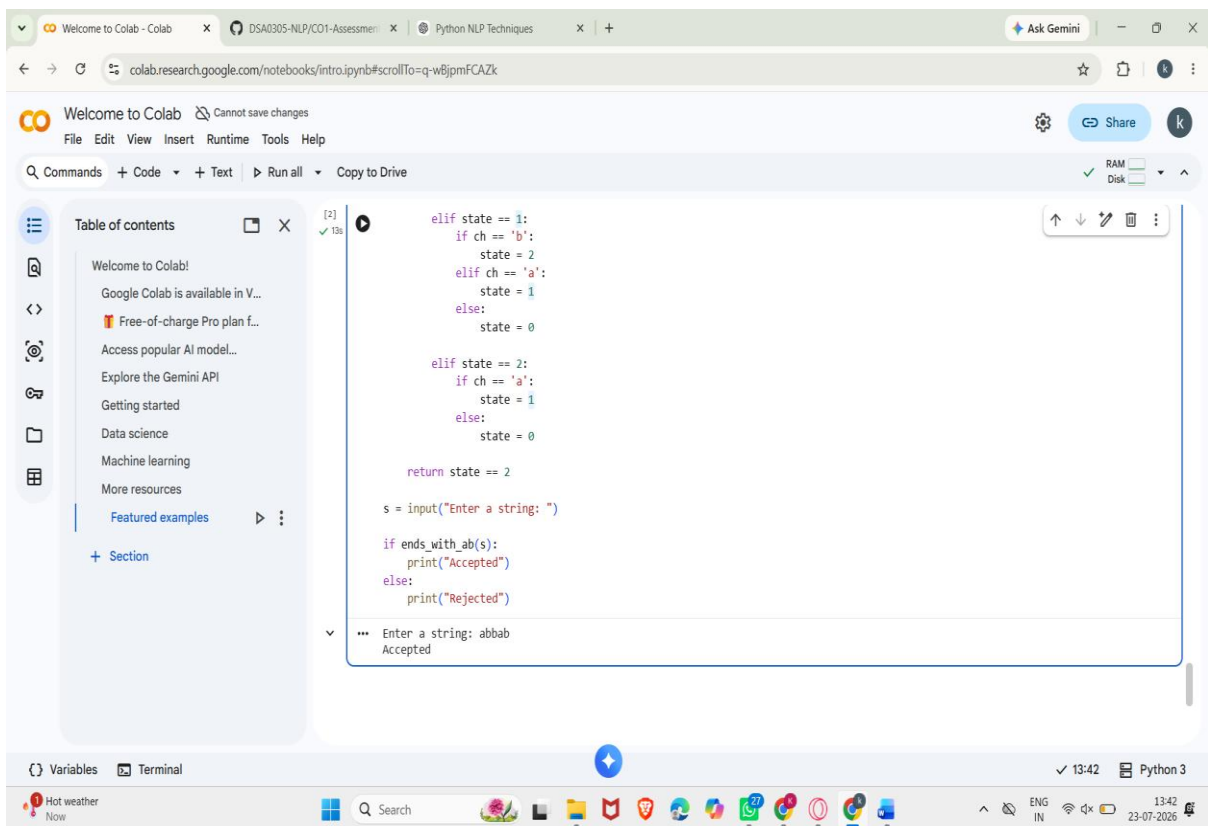
    print("Accepted")

else:

    print("Rejected")

```

Output



The screenshot shows a Google Colab notebook interface. The top bar includes the Google Colab logo, a 'Welcome to Colab' message, and a 'Share' button. The left sidebar contains a 'Table of contents' panel with links to 'Welcome to Colab!', 'Google Colab is available in V...', 'Free-of-charge Pro plan f...', 'Access popular AI model...', 'Explore the Gemini API', 'Getting started', 'Data science', 'Machine learning', and 'More resources'. The main area displays a Python script with the following code:

```

elif state == 1:
    if ch == 'b':
        state = 2
    elif ch == 'a':
        state = 1
    else:
        state = 0

elif state == 2:
    if ch == 'a':
        state = 1
    else:
        state = 0

return state == 2

s = input("Enter a string: ")

if ends_with_ab(s):
    print("Accepted")
else:
    print("Rejected")

```

Below the code, the output of the script is shown:

```

... Enter a string: abbab
Accepted

```

The bottom of the interface shows a 'Variables' panel, a 'Terminal' panel, and a system tray with the date and time (13:42, 23-07-2026).

